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Complex reaction processes in combustion unraveled by neural network-based molecular dynamics simulation

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Combustion is a complex chemical system which involves thousands of chemical reactions and generates hundreds of molecular species and radicals during the process. In this work, a neural network-based molecular dynamics (MD) simulation is carried out to simulate the benchmark combustion of methane. During MD simulation, detailed reaction processes leading to the creation of specific molecular species including various intermediate radicals and the products are intimately revealed and characterized. Overall, a total of 798 different chemical reactions were recorded and some new chemical reaction pathways were discovered. We believe that the present work heralds the dawn of a new era in which neural network-based reactive MD simulation can be practically applied to simulating important complex reaction systems at ab initio level, which provides atomic-level understanding of chemical reaction processes as well as discovery of new reaction pathways at an unprecedented level of detail beyond what laboratory experiments could accomplish.

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Ever since learning to use fire, human beings have never stopped studying combustion. With increasingly serious concern on environmental pollution from combustion, understanding and mastering the combustion mechanisms is of great importance. Gaining fundamental insights into combustion processes can help us design more efficient engines and minimize the production of pollutants. A typical combustion may contain hundreds of chemical species and thousands of fundamental chemical reactions. In particular, combustion occurs at extreme physical conditions with high pressures and high temperatures up to several thousand degrees. Also, many elementary reactions in a combustion typically occur on sub picosecond time scale. These extreme physical conditions make it very difficult, if not impossible, to carry out real-time experimental study of combustion. Thus, most experimental investigations of chemical reaction mechanisms focus on individual reactions instead of the complex reaction processes occurring in a combustion. In the past decades, in slico experiments such as reactive molecular dynamics (MD) simulations have shown their values in providing molecular (atomic)-level insights into the mechanism of combustions. In a reactive MD simulation, the reaction condition can be easily controlled in the simulation and some supercritical conditions that are difficult to achieve in the experiment can also be handled. Compared with the traditional theoretical approaches such as transition sate theory and quantum collision theory that focuses on studying a single reaction, reactive MD simulation can construct the entire interwoven reaction network of a combustion system[1]. The heart of the reactive MD simulation is the potential energy surface (PES), which describes the inter- and intramolecular interactions for molecules. Currently, there are mainly two classes of methods that can be used to construct the PES of a given molecular system: the quantum mechanics (QM)-based methods and the empirical force fields. Quantum mechanics is undoubtedly more rigorous and accurate, and MD simulations based on it are known as ab initio MD simulation (AIMD)[2,3]. Although the AIMD method in principle can simulate complex chemical reactions in real time, it is limited to relatively small systems and short simulation time (typically, dozens of picoseconds) due to exorbitant computational costs of on-the-fly ab initio calculation. With the rapid development of computer hardware and algorithms, especially the employment of graphic processing units (GPUs), some AIMD methods have recently begun to handle larger chemical systems[4]. But so far, it is still impractical to use AIMD to simulate large-scale complex reaction systems such as combustions. Over the past decades, many reactive force fields (or PESs) have been developed and successfully used for various reactive molecular systems[5–12]. A comprehensive discussion of these reactive force fields can be found in refs. [13,14]. Among these force fields, the empirical ReaxFF was widely used in MD simulation of combustion systems due to its computational efficiency[15], but its accuracy and reliability are of significant concern[16–18]. The key points of developing a reaction force field are the choice of the functional form and the parameterization process, which are complicated and depend on human intervention.

Recently, more researchers are switching to seek the help of machine-learning (ML) methods. ML method, especially artificial neural networks (NN), provides the possibility to construct PESs with the accuracy of the QM method but with an efficiency comparable to that of force fields. Neural networks constitute a very flexible and unbiased class of mathematical functions, which in principle is able to approximate any real-valued function to arbitrary accuracy. Since Behler and Parrinello proposed the high-dimensional neural network approach[19,20], several methods have been developed to implement this approach and many different kind of NN PESs have been proposed for water, small organic molecules, and metalloid materials[21–25]. For example, the sGDML[26–28], SchNet[29], PhysNet[30], and FCHL[31] methods. NN potentials have also been employed to study the reaction mechanisms of chemical systems. By combining high-precision NN PESs and quantum collision theory, Zhang and Jiang's group have studied a series of elementary reactions in the gas phase and on the surface[32–35]. Liu and co-workers developed the LASP program to study the heterogeneous catalysis with NN PESs[36] and built stochastic surface walking (SSW)-NN to explore reaction pathways from glucose to 5-hydroxymethylfurfural[37]. Brickel et al. also studied the nucleophilic substitution reaction $[Cl–CH_3–Br]^-$ in water with NN potential[38].

In this report, we present an in silico simulation of methane combustion based on an NN potential derived by training a high-dimensional NN model from ab initio computed energies. To achieve high efficiency and accuracy, the DeePMD model was used[39–41]. This NN PES can accurately predict the energy and atomic forces of reactants, products and reaction intermediates. Based on this model, a 1-ns reactive MD simulation was performed for a combustion system initially containing 100 methane and 200 oxygen molecules with a sub-femtosecond time resolution (Fig. 1). A complete reaction network of the methane combustion can be constructed from the MD trajectory. The simulation not only produced the main reaction pathways that are consistent with the experiment but also provided much more detailed insights about the combustion processes as will be described in the following.

Results

Accuracy of the NN PES. The performance of the NN potential highly depends on the quality of the reference datasets. Although several databases, such as QM7[42], QM9[43], ANI-1[44], and ANI-1x[45], are accessible, they mainly include organic molecules and are therefore not suitable for this work. Combustion of methane will generate many molecular fragments and a lot of them are free radicals[46]. Therefore, we followed a workflow (details are listed in the "Methods" section) to construct the reference datasets for the combustion. Then the DeepPot-SE model[47] was used to train the NN PES based on the reference. The predictive power of the NN model is shown in Supplementary Table 1 and Supplementary Fig. 1. It is clear that the DFT energies can be accurately reproduced by the NN model. The mean absolute errors are only 0.04 and 0.14 eV/atom in the training set and the test set, respectively. As for the atomic forces, the predicted values of the NN model are also highly consistent with the calculated results of the DFT (Supplementary Fig. 1). The correlation coefficient is 0.999 and the MAE is 0.12 eV/Å. Considering that there are a large number of atomic and molecular collisions during the combustion process, and some atomic forces can be as high as dozens of eV/Å, the accuracy of the NN model is encouraging. To verify the energy conservation of the NN PES, we performed a reactive MD simulation under the NVE ensemble. The system is a periodic box containing 100 CH_4 molecules and 200 O_2 molecules (a total of 900 atoms) with a density of 0.25 g/cm^3. As shown in Supplementary Fig. 2, the total energy is conserved in MD simulation.

The initial stage of combustion. A 1 ns reactive MD simulation was performed for methane combustion with the NN PES under the NVT ensemble. The system is also a periodic box containing 100 CH_4 molecules and 200 O_2 molecules (a total of 900 atoms) with a density of 0.25 g/cm^3. The MD simulations were run with a time-step of 0.1 fs and the temperature was kept at 3000 K by using the Berendsen thermostat. We chose a relatively high density (and thus high pressure) and high temperature to